

2.2 Carbon Cycle & Energy Security

Geography — Eduqas A-Level (A110QS) — Component 2: Global Systems

1. The Global Carbon Cycle

The **carbon cycle** is the biogeochemical circulation of carbon between four great Earth reservoirs: the *lithosphere* (rocks and sediments), the *hydrosphere* (oceans), the *atmosphere* (as CO₂ and CH₄) and the *biosphere* (living and dead organic matter). Like the water cycle, it is a *closed system* at planetary scale — carbon is conserved — but it couples *two interacting sub-cycles* operating on utterly different timescales: a **fast biological cycle** that turns over in years to centuries, and a **slow geological cycle** that turns over in millions of years. Anthropogenic fossil-fuel combustion is currently transferring carbon from the slow store to the fast cycle at 100–200 times the natural geological rate, and this asymmetry is the root cause of anthropogenic climate change.

Carbon cycle: The biogeochemical cycle by which carbon is exchanged between the lithosphere, hydrosphere, atmosphere and biosphere through processes including photosynthesis, respiration, decomposition, weathering, ocean exchange, combustion and volcanism. Globally closed; the mass of carbon is conserved but continuously redistributed between stores.

Global Carbon Stores (Reservoirs)

Carbon is distributed very unequally across Earth's reservoirs. The sedimentary lithosphere holds roughly 100 million GtC (gigatonnes of carbon), over 99% of all planetary carbon, but this is immobile on human timescales. The smaller, dynamic reservoirs — ocean, atmosphere, biosphere, soils, fossil fuels — together total ~46,000 GtC and are the ones that respond to human activity. Two observations matter for exam answers: (i) the atmosphere is a *small store* (~870 GtC) so relatively small flux imbalances produce large % changes, and (ii) soils store *more carbon than vegetation plus the atmosphere combined*, so land-use change matters disproportionately.

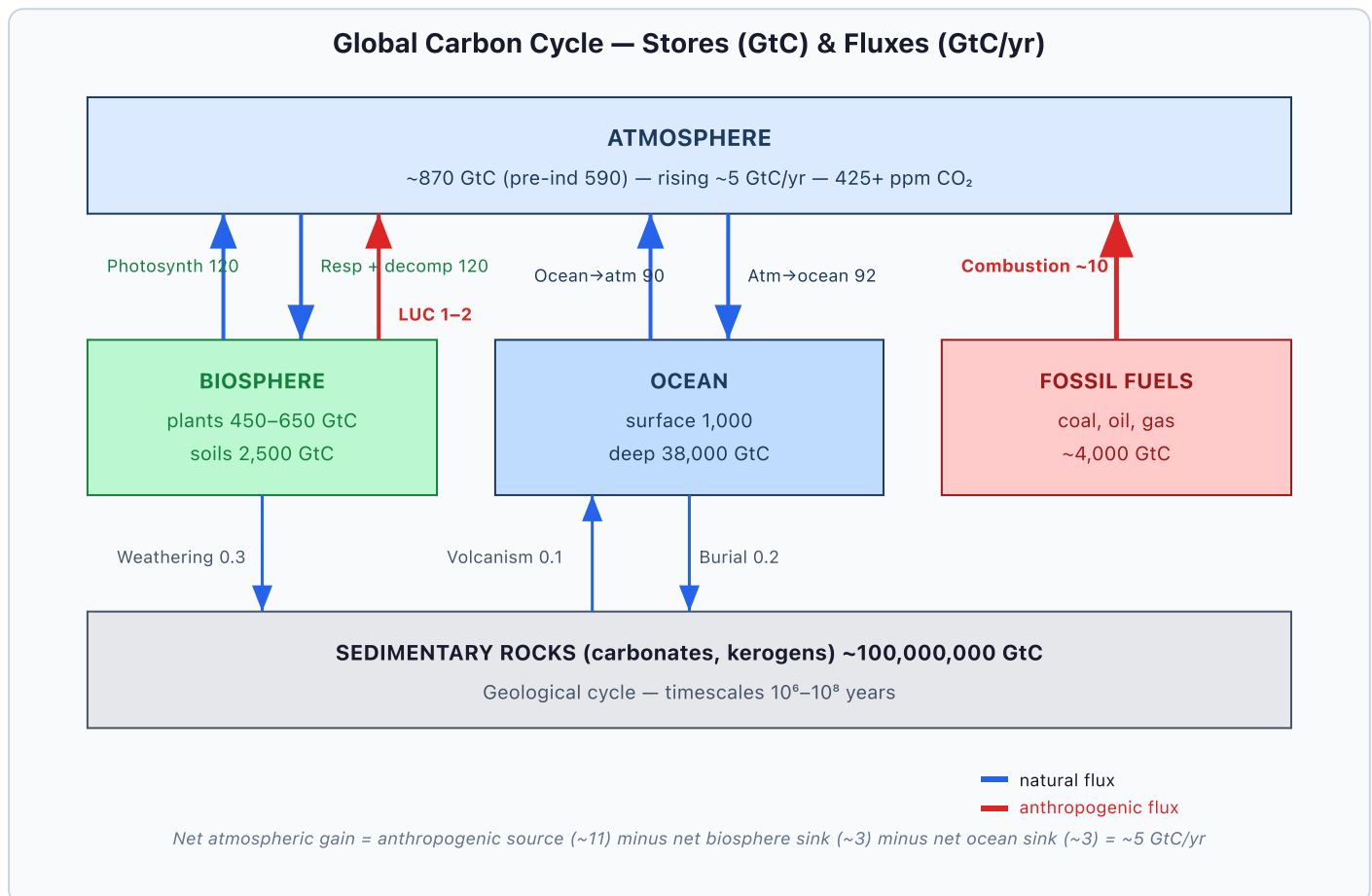
Store	Approximate size (GtC)	Share of dynamic C	Notes
Sedimentary carbonates (limestone, dolomite, shales)	~100,000,000	— (geological)	Largest store; locked for 10^6 – 10^8 years
Deep ocean (dissolved inorganic C)	~38,000	~83%	Bicarbonate HCO_3^- dominates
Fossil fuels (coal, oil, gas)	~4,000	~9%	Finite; subject to anthropogenic extraction
Soils (top 1 m, including peat)	~2,500	~5%	Exceeds atmosphere + biosphere combined
Surface ocean (0–100 m)	~1,000	~2%	Exchanges rapidly with atmosphere
Atmosphere (CO_2 , CH_4)	~870 (pre-ind ~590)	~2%	425+ ppm CO_2 in 2024
Biosphere (plants + animals)	~450–650	~1%	Forests hold ~80% of terrestrial biomass C
Permafrost (northern high latitudes)	~1,500 (soil C, within the 2,500 above)	—	Major tipping-point risk if thawed

Carbon reservoir (store): A natural sink or repository holding carbon in a particular form for a characteristic residence time. Reservoirs are connected by *fluxes* — the flows of carbon between them, measured in GtC/yr. Residence times vary from days (atmospheric CO_2 against biosphere) to hundreds of millions of years (sedimentary carbonates).

Global Carbon Fluxes

Natural fluxes between reservoirs are large but essentially balanced over the Holocene: photosynthesis and respiration each move ~120 GtC/yr between atmosphere and biosphere; ocean–atmosphere exchange moves ~90 GtC/yr in each direction. The tiny slow-cycle fluxes of volcanism (~0.1 GtC/yr) and

silicate weathering (~ 0.3 GtC/yr) regulate atmospheric CO_2 on geological timescales. The anthropogenic flux of ~ 10 GtC/yr from fossil-fuel combustion plus $\sim 1\text{--}2$ GtC/yr from land-use change is *small* compared with photosynthesis, but it is an *imbalance*: the biosphere and oceans together reabsorb only about half of it, leaving ~ 5 GtC/yr accumulating in the atmosphere. That ~ 5 GtC/yr is the cause of the observed ~ 2.4 ppm/yr rise in atmospheric CO_2 .



Exam tip — know three flux numbers cold: A strong answer always anchors itself with quantified fluxes. Memorise (i) photosynthesis / respiration ~ 120 GtC/yr, (ii) anthropogenic fossil combustion ~ 10 GtC/yr, and (iii) atmospheric accumulation ~ 5 GtC/yr. The ratio of anthropogenic to natural exchange (10 : 120) looks small but is a *net perturbation* because the 120 figure is nearly balanced.

The Atmospheric CO_2 Budget

A standard way to think about the anthropogenic carbon flux is as a *budget*: sources on one side, sinks on the other, and any imbalance accumulating in the atmosphere. The Global Carbon Project publishes this annually. For 2023 (latest full year):

- **Sources:** fossil CO₂ 10.2 GtC + land-use change ~1.3 GtC = ~11.5 GtC
- **Sinks:** ocean ~2.9 GtC + land ~3.5 GtC = ~6.4 GtC (the “natural subsidy”)
- **Atmospheric growth:** ~5.1 GtC (equivalent to ~2.4 ppm rise)

The natural sinks have absorbed about 55% of all cumulative anthropogenic emissions since 1850. This is a vast ecosystem service — without it, atmospheric CO₂ would already exceed 550 ppm. But the *fraction* absorbed is likely to decline as oceans warm and forests saturate, meaning *the same emissions will cause more warming in future* — a crucial, often-missed point in exam answers.

Methane (CH₄) — the Forgotten Carbon

Although Topic 2.2 centres on CO₂, methane is the second-largest anthropogenic greenhouse gas with a global-warming potential of ~82 over 20 years and ~30 over 100 years. Atmospheric CH₄ has risen from ~720 ppb (pre-industrial) to ~1,930 ppb in 2024 — a 168% rise, far larger proportionally than CO₂. Sources: enteric fermentation in ruminants (~30%), oil and gas leaks (~20%), landfill (~15%), rice paddies (~10%), biomass burning. The *Global Methane Pledge* (Glasgow COP26, 2021) aims to cut anthropogenic methane 30% by 2030 — the single most effective short-term climate action, because CH₄'s short atmospheric lifetime (~9 years) means cuts feed rapidly into reduced warming.

2. Short-Term (Biological) vs Long-Term (Geological) Cycles

The carbon cycle can only be understood as the superposition of two cycles of vastly different tempo. Candidates should be able to compare both in A-level AO1 questions.

The Short-Term (Biological / Fast) Cycle

The biological cycle operates over hours to centuries and transfers carbon between the atmosphere, biosphere and surface ocean.

- **Photosynthesis:** plants and phytoplankton use solar energy to convert atmospheric CO₂ + H₂O into glucose (C₆H₁₂O₆) + O₂, fixing ~120 GtC/yr on land and ~50 GtC/yr in the surface ocean.
- **Respiration:** plants, animals and microbes oxidise organic matter back to CO₂, returning ~120 GtC/yr.
- **Decomposition:** saprotrophic bacteria and fungi break down dead organic matter. In warm, aerobic conditions this returns carbon rapidly as CO₂; under anaerobic conditions (e.g. waterlogged peat)

decomposition is partial, producing CH_4 and preserving carbon in soils.

- **Ocean–atmosphere exchange:** CO_2 dissolves in surface water according to Henry's law, then dissociates into bicarbonate and carbonate ions (the *carbonate buffer*).

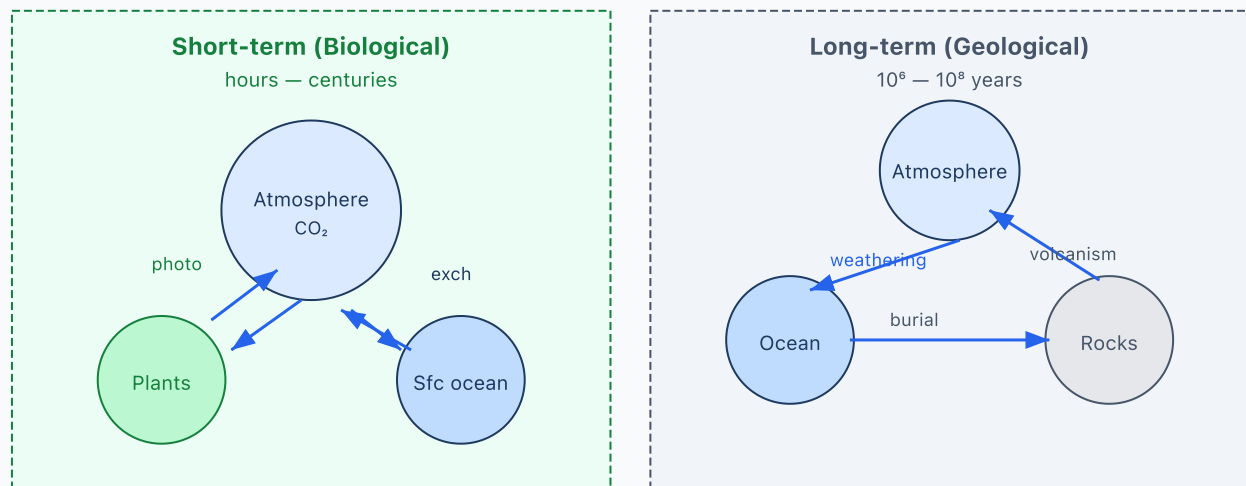
The Long-Term (Geological / Slow) Cycle

The geological cycle operates over 10^6 – 10^8 years and transfers carbon between rocks, oceans and atmosphere through weathering, sedimentation and tectonics.

- **Silicate weathering:** atmospheric CO_2 dissolves in rainwater ($\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3$) and reacts with silicate minerals (CaSiO_3), releasing Ca^{2+} and HCO_3^- ions to the oceans. This *consumes* CO_2 and is the dominant long-term sink.
- **Carbonate deposition:** marine organisms (corals, foraminifera, coccolithophores) precipitate calcium carbonate (CaCO_3) shells. Sinking shells accumulate as limestone.
- **Burial:** organic matter buried in anoxic sediments becomes kerogen and, over millions of years under heat and pressure, coal, oil or gas.
- **Subduction and metamorphism:** sea-floor carbonate rocks descend into the mantle; heating releases CO_2 which returns to the atmosphere via volcanism.
- **Volcanic outgassing:** $\sim 0.1 \text{ GtC/yr}$ from arc volcanoes and mid-ocean ridges. Over geological time this replaces the weathering sink; the balance sets Earth's long-term CO_2 level and habitability.

Exam tip — contrast the two timescales: Questions that ask you to "explain the operation of the carbon cycle" score top bands when candidates explicitly compare fast (biological) and slow (geological) components and note that *human activity is short-circuiting the slow cycle into the fast one* — releasing in 200 years what took 100 million years to bury.

Short-term (Biological) vs Long-term (Geological) Carbon Cycle

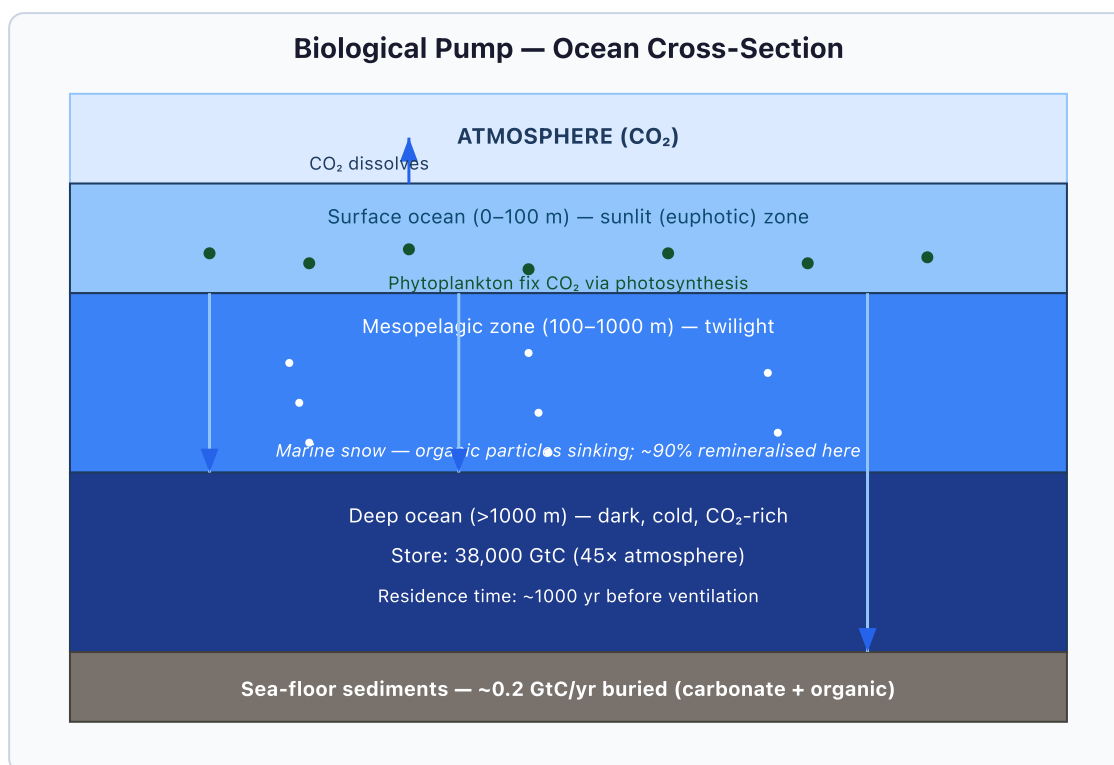


3. The Biological (Ocean) Pump

The **biological pump** is the ocean's most important long-term carbon sink. It has three linked steps: photosynthesis by surface phytoplankton fixes dissolved CO₂ into organic carbon; a fraction of that carbon sinks out of the sunlit zone as particles (faecal pellets, dead cells, aggregates) popularly called *marine snow*; and a small fraction (~0.2 GtC/yr) is buried in sea-floor sediments, removed from the active cycle for millions of years. Without the biological pump, atmospheric CO₂ would be ~200 ppm higher.

Biological pump: The sequence of biological and gravitational processes by which surface-ocean photosynthesis transfers carbon from the atmosphere to the deep ocean and ultimately to marine sediments. It comprises the *soft-tissue pump* (organic matter sinking) and the *carbonate pump* (CaCO₃ shell sinking); together they maintain a vertical gradient of ~15% more dissolved inorganic carbon at depth than at the surface.

Complementing the biological pump is the **physical (solubility) pump**: CO₂ is more soluble in cold water, so cold polar surface waters absorb CO₂ before sinking in the North Atlantic and around Antarctica. Together the two pumps explain why the deep ocean holds ~38,000 GtC — 45 times the atmospheric store.



Exam tip — distinguish the two pumps: Examiners reward candidates who *separate* the biological pump (photosynthesis + sinking organic matter) from the solubility (physical) pump (cold water dissolving more CO₂). Both together explain why the deep ocean is the dominant carbon sink. Warming weakens the solubility pump, and ocean acidification may weaken the carbonate pump — both are positive feedbacks.

4. Human Disruption of the Carbon Cycle

Since ~1750 human activity has released an estimated ~2,400 GtCO₂ (~650 GtC), roughly half of which remains in the atmosphere. The principal anthropogenic flux pathways are fossil-fuel combustion, cement manufacture, deforestation and other land-use change.

Fossil-Fuel Combustion

Burning coal, oil and natural gas converts geological carbon to atmospheric CO₂ almost instantaneously. Global fossil CO₂ emissions reached a record ~37.4 Gt CO₂ in 2023 (~10.2 GtC). China is the largest emitter (~30% of global total); the USA (~13%); the EU (~7%); India (~7%). On a *per-capita* basis, however, the USA, Australia and Gulf states remain several times above the global average, and historic cumulative emissions remain dominated by the Global North.

Cement Production

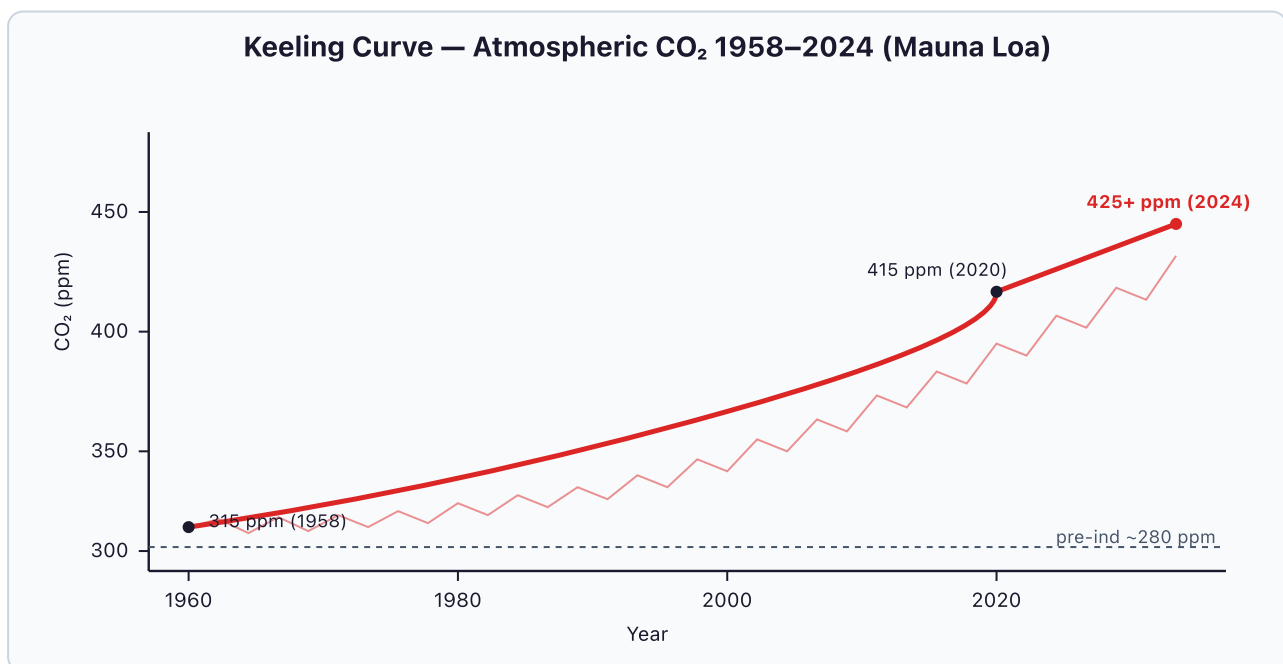
Cement is manufactured by calcining limestone: $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$. This *process emission* alone accounts for ~5% of global CO_2 , not including the fuel burnt in the kiln. Cement is thus a "hard-to-abate" industrial sector and a major target for carbon capture.

Deforestation and Land-Use Change

Clearing tropical forest releases biomass carbon directly (burning) and soil carbon gradually (oxidation of disturbed soils). The net land-use-change flux is ~1–2 GtC/yr, dominated by the Amazon, the Congo basin and tropical Southeast Asia (Indonesia's peat-swamp forests are especially carbon-dense). Ploughing prairies and converting peatlands also releases soil organic carbon. Regrowth on abandoned land offsets some of this loss.

Atmospheric CO_2 and the Keeling Curve

The Mauna Loa CO_2 record, begun by Charles Keeling in 1958, is the single most iconic scientific curve of the Anthropocene. From 315 ppm in 1958, CO_2 passed 400 ppm in 2013 and exceeded 425 ppm in 2024 — higher than at any time in at least ~3 million years (mid-Pliocene). The curve has a sawtooth annual cycle (~6–7 ppm amplitude) reflecting Northern Hemisphere growing-season uptake, superimposed on a relentless rising trend that has accelerated from ~0.9 ppm/yr in the 1960s to ~2.4 ppm/yr today.



Ocean Acidification

Roughly a quarter of anthropogenic CO₂ has been absorbed by the oceans, moderating the atmospheric rise but shifting the carbonate equilibrium:



The added H⁺ ions have lowered ocean surface pH by ~0.1 units since the industrial era (from ~8.2 to ~8.1), a 30% increase in acidity on the logarithmic scale. A further 0.3–0.4 drop is projected by 2100 under high-emission pathways. The falling carbonate ion concentration threatens *calcifying organisms* (corals, pteropods, coccolithophores) whose shells become energetically more costly to build and ultimately dissolve when waters become undersaturated with respect to aragonite or calcite.

Ocean acidification: The ongoing decrease in seawater pH and carbonate-ion concentration caused by uptake of anthropogenic CO₂. Since 1750 surface-ocean pH has fallen by ~0.1, a 30% increase in H⁺ concentration. It threatens calcifying organisms and coral-reef ecosystems, and may reduce the ocean's future capacity as a carbon sink.

Exam tip — acidification ≠ warming: Ocean acidification is a *separate consequence* of CO₂ emissions from global warming. Both share the same cause but act through different mechanisms. Always mention this distinction if a question asks about impacts of rising CO₂ — it shows analytical depth.

5. Feedbacks and Tipping Points

The carbon–climate system contains many **feedback loops**: responses in which an initial change drives a secondary change that either amplifies (positive feedback) or dampens (negative feedback) the original perturbation. Feedbacks determine how sensitively climate responds to anthropogenic forcing, and several are believed to operate around *tipping points* — thresholds beyond which change becomes self-sustaining and irreversible on human timescales.

Feedback loop: A process by which the output of a system modifies its own input. A *positive (amplifying) feedback* reinforces the original change; a *negative (stabilising) feedback* opposes it.

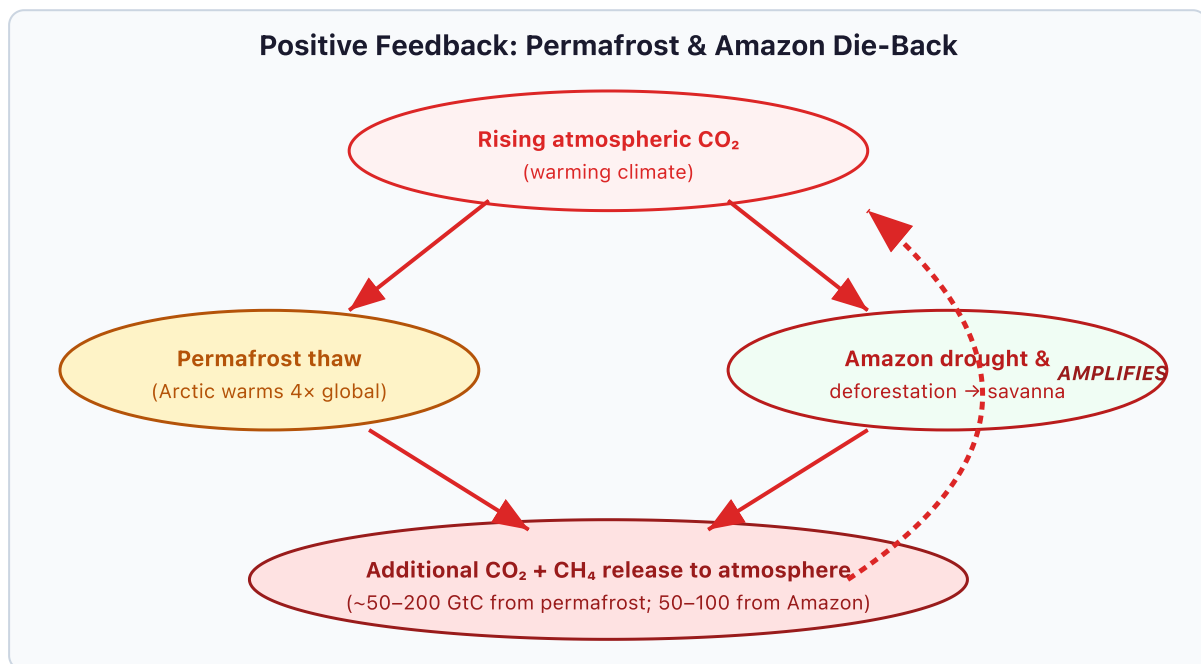
Positive carbon-cycle feedbacks such as permafrost thaw are of central concern because they could convert a rising-temperature "nudge" into runaway warming.

Feedback	Sign	Mechanism	Magnitude (to 2100)
Permafrost thaw	+(amplifying)	Frozen soil organic matter decomposes to CO ₂ /CH ₄ when thawed	~50–200 GtC additional release
Amazon die-back	+	Warming + drought + deforestation flips forest to savanna; releases stored C	~50–100 GtC if triggered
Ice–albedo	+	Melting sea/land ice lowers albedo → more absorbed heat → more melt	Indirect via warming
Wildfire increase	+	Hotter, drier conditions raise fire frequency; burns release stored vegetation C	Regional; boreal & tropical forests
Ocean outgassing	+	Warmer water holds less CO ₂ (Henry's law); weaker biological + solubility pumps	Reduced ocean sink fraction
CO ₂ fertilisation	–(stabilising)	Higher CO ₂ enhances plant photosynthesis and growth	Partial; limited by N, P, water
Silicate weathering	–	Warmer, wetter climate speeds weathering → consumes atmospheric CO ₂	Only significant on 10 ⁵ + yr timescales
Ocean initial uptake	–	Higher atmospheric CO ₂ increases dissolution into surface waters	Already absorbing ~25% of emissions

Major Tipping Points

- **Amazon die-back:** beyond ~20–25% deforestation combined with warming, reduced evapotranspiration could flip the eastern Amazon from rainforest to savanna. Gatti et al. (2021) already show parts of the SE Amazon are a net CO₂ *source*.
- **Permafrost collapse:** abrupt thaw via thermokarst could release carbon stored for 10,000+ years, with ~1,500 GtC at risk across Arctic permafrost.

- **AMOC weakening:** freshening of the North Atlantic from Greenland melt may slow the Atlantic Meridional Overturning Circulation, cooling Western Europe but shifting tropical rainfall belts.
- **Arctic summer sea ice loss:** likely before 2050 on current trajectory; amplifies Arctic warming and may weaken the polar vortex.
- **Coral reef collapse:** combined warming and acidification beyond 1.5–2°C threatens most tropical reefs; GBR mass-bleaching years 2016, 2017, 2020, 2022, 2024.



Exam tip — always classify the feedback: When you mention a feedback, immediately label it positive/negative, name the mechanism, and ideally give a magnitude. "Permafrost thaw is a positive feedback: rising temperatures thaw frozen soils, releasing 50–200 GtC as CO₂ and CH₄, which further warms the climate" is a model AO1/AO2 sentence.

5b. Climate Change — Observed and Projected Impacts

The carbon cycle's disruption manifests as climate change. By 2024 global mean surface temperature had risen ~1.3°C above the 1850–1900 pre-industrial baseline, with 2023 and 2024 the two hottest years on record, briefly exceeding 1.5°C in individual months. Key physical impacts include:

- **Sea-level rise:** ~21–24 cm since 1880; currently ~4.6 mm/yr (accelerating). IPCC AR6 projects 0.44–0.76 m by 2100 under SSP2-4.5; high-end ice-sheet scenarios 2 m+ by 2100 cannot be ruled

out.

- **Cryospheric decline:** Arctic sea-ice summer minimum down ~13%/decade; Greenland losing ~280 Gt ice/yr; Antarctica ~150 Gt/yr (accelerating West Antarctic Ice Sheet).
- **Heatwaves:** 2003 European (70,000 excess deaths), 2010 Russian, 2019 European, 2021 Pacific Northwest (49.6°C in Lytton, BC), 2022 UK (40.3°C at Coningsby — first UK 40°C), 2024 India/Pakistan.
- **Shifting precipitation:** intensification of the hydrological cycle per Clausius–Clapeyron (~7% more atmospheric moisture per °C). Wet regions wetter, dry regions drier; monsoon disruption.
- **Ecosystem disruption:** species range shifts polewards ~17 km/decade; coral bleaching; insect phenology mismatches.
- **Agriculture:** yield gains from CO₂ fertilisation offset by heat and water stress; wheat yields projected to fall in low latitudes.

Exam tip — connect carbon cycle to climate impact: Eduqas favours candidates who explicitly join the chain: human disruption of the C cycle → rising atmospheric CO₂ → enhanced greenhouse effect → physical climate change → ecological/social/economic consequences. Show all five links.

6. Energy Security and Energy Mix

The second half of this topic shifts from biogeochemistry to **energy geography**. The two halves are tightly linked: the dominant anthropogenic driver of the carbon cycle is energy-related emissions, so decisions about energy supply *are* decisions about climate.

Energy security: The uninterrupted availability of energy sources at an affordable price for household, industrial and transport needs. Includes three dimensions — *availability* (physical supply), *accessibility* (geopolitical/economic reach) and *affordability* (price stability). Threats include resource depletion, geopolitical disruption, infrastructure attack and climate policy transition.

Energy mix: The combination of primary energy sources (coal, oil, natural gas, nuclear, hydro, wind, solar, bioenergy, geothermal) used to meet total energy demand in a country or region.

National energy mixes reflect resource endowment, level of development, policy choices and historical path dependency.

Primary vs Final Energy

Careful terminology matters. **Primary energy** is the raw form — the chemical energy in a tonne of coal or the kinetic energy in a wind gust. **Final (end-use) energy** is what reaches the consumer after conversion and transmission losses. A coal power station delivers only ~40% of primary energy as electricity; wind and solar, which "skip" the thermal step, have much higher primary-to-final ratios. This is why decarbonisation scenarios typically *reduce* total primary energy demand even as service provision grows.

National Energy Mixes 2024

Country	Electricity mix (approx.)	Key feature
UK	45% renewables (wind dominant), 30% gas, 14% nuclear, <1% coal	Coal phase-out completed Sept 2024; Hinkley Point C under construction
USA	42% natural gas, 18% coal, 19% nuclear, 21% renewables	Shale-gas revolution since 2008; Inflation Reduction Act 2022
France	~70% nuclear, ~20% renewables (hydro/wind), ~6% gas	Lowest-carbon grid in G7; new EPR reactors (Flamanville)
Germany	~55% renewables, ~25% coal/lignite, ~15% gas, 0% nuclear (2023)	Energiewende; nuclear shutdown; scramble to replace Russian gas
Norway	~92% hydro, ~7% wind, <1% thermal	Europe's green battery; net oil/gas exporter
China	~58% coal, ~30% renewables (huge installed solar/wind/hydro), 5% nuclear, gas	Largest emitter but largest renewables build-out globally
India	~70% coal, ~22% renewables + hydro, ~3% nuclear	Rapid demand growth; 2070 net-zero target

Saudi Arabia	~99% fossil fuels (oil + gas)	Vision 2030 plans 50% renewables by 2030 but slow progress
Brazil	~55% hydro, ~20% wind/solar, ~12% bioenergy (sugar-cane ethanol), fossil	Amazon hydro; leading ethanol economy

Exam tip — compare two contrasting countries: 9–20 mark Eduqas essays on energy often ask candidates to "compare" national energy profiles. Pair one fossil-heavy state (Saudi Arabia, Kuwait) with one renewables leader (Norway, France or Iceland) and draw out *why* — resource endowment, colonial history, state capacity, policy choice.

Geopolitics of Energy

Energy has always been geopolitical. The 1973 OPEC oil embargo quadrupled prices and triggered Western stagflation; the 1979 Iranian Revolution did so again; the 1990 Iraq invasion of Kuwait spurred the First Gulf War; gas cut-offs from Russia to Ukraine (2006, 2009, 2014) foreshadowed the 2022 invasion. Russia's weaponisation of gas in 2022 — piping ~40% of EU gas pre-war — exposed fossil interdependence as a strategic vulnerability and accelerated European renewables and efficiency deployment. The 2020s see a reconfiguration: China is the largest energy importer; the USA has become a net energy exporter (LNG) for the first time since 1952; the Gulf states hedge between oil exports and renewables diplomacy (UAE hosted COP28).

Energy Poverty and the Just Transition

Roughly 760 million people lack electricity access (IEA 2024, mostly sub-Saharan Africa) and 2.3 billion rely on polluting cooking fuels (wood, charcoal, dung) causing ~3 million indoor-pollution deaths/yr. An equitable transition must expand clean energy to the unconnected while decarbonising the over-connected. The concept of *climate justice* recognises asymmetry: the poorest 50% globally are responsible for ~12% of emissions but suffer ~75% of climate impacts. A *just transition* for workers in fossil-dependent regions (Appalachia coal, Alberta oil sands, West Yorkshire, Silesia) demands retraining, investment, social support — or it risks political backlash that derails mitigation.

7. Fossil Fuels — Conventional and Unconventional

Fossil fuels — **coal, oil and natural gas** — still supply ~80% of global primary energy. They are compact, transportable, on-demand sources with mature infrastructure, and underpin petrochemicals (plastics, fertilisers, pharmaceuticals). But they are finite, geographically uneven, and generate climate, air-pollution and ecological externalities.

Conventional Reserves

- **Coal:** largest reserves in the USA, Russia, Australia, China, India. Highest CO₂ per unit energy (~95 g CO₂/MJ). Responsible for ~35–40% of global electricity and ~40% of fossil CO₂.
- **Oil:** dominated by the Persian Gulf (Saudi Arabia, Iraq, Iran, UAE, Kuwait), Russia, USA and Venezuela. Refined into petrol, diesel, jet fuel, heating oil, naphtha. ~33% of global primary energy.
- **Natural gas:** Russia, Iran, Qatar, USA, Turkmenistan hold ~65% of reserves. Lower carbon intensity per MJ (~55 g/MJ) but leakage of methane (GWP ~82 over 20 yr) can erase this advantage.

Unconventional Reserves

When conventional reservoirs deplete, the industry turns to *unconventional* sources — harder to extract, costlier, more environmentally intrusive:

- **Shale gas (fracking):** natural gas trapped in low-permeability shale, extracted by horizontal drilling and hydraulic fracturing. Transformed US energy since 2008 (Marcellus, Permian) but blamed for methane leakage, water contamination, induced seismicity.
- **Tar (oil) sands:** bituminous sands mined and upgraded. Alberta, Canada, holds the third-largest oil reserves globally. Energy-intensive extraction; emissions 15–20% higher than conventional oil.
- **Deep-water and ultra-deep-water oil:** Gulf of Mexico, Brazilian pre-salt (Santos basin), West Africa. Brings Deepwater Horizon-type risks.
- **Arctic oil and gas:** see Section 16.
- **Coalbed methane** and **tight oil:** minor but growing unconventional streams.

Fracking (hydraulic fracturing): An unconventional-gas technique in which high-pressure fluid (water, sand, chemicals) is injected into shale rock to create fractures that release trapped gas. Controversial for methane leakage, groundwater-contamination risks, induced seismicity and surface-land disturbance; banned in France, Germany, New York; moratoria in UK and Scotland.

Environmental and Social Costs

- CO₂ and CH₄ emissions drive climate change.
- Air pollution (PM_{2.5}, NO_x, SO₂) kills ~4–8 million people annually.
- Ocean oil spills (Deepwater Horizon 2010, Exxon Valdez 1989) damage ecosystems for decades.
- Landscape destruction from mountaintop-removal coal mining (Appalachia).
- Water use for fracking and tar sands.
- Gas flaring in the Niger Delta, Permian Basin.
- Geopolitical instability: "oil curse", rentier states, pipeline wars.

8. Renewable Energy

Renewable sources (including hydro) have grown from ~18% of global electricity in 2000 to over 30% in 2024, driven by collapsing costs: utility solar PV fell from ~\$6/W in 2008 to ~\$0.30/W in 2024, and onshore wind LCOE is now typically the cheapest new-build electricity. The IEA expects renewables to exceed coal in global generation around 2025.

Source	Typical capacity factor	LCOE 2024 (\$/MWh)	Strengths	Constraints
Solar PV (utility)	15–25%	30–50	Modular, silent, falling cost	Intermittent; no night output; land area
Onshore wind	25–40%	30–55	Cheap, mature, low land footprint	Intermittent; visual/noise impacts
Offshore wind	40–55%	70–100	Steadier winds; out of sight	High capital cost; grid connection
Hydro (large)	35–55%	30–70	Dispatchable; huge scale	Displacement; methane from reservoirs; sediment
Geothermal	70–95%	50–100	Baseload; small footprint	Site-specific (Iceland, Kenya, Indonesia)

Biomass	50–80%	80–130	Dispatchable; uses waste	Land competition; supply-chain C; air pollution
Tidal barrage/stream	20–35%	150–250	Predictable; two-tide daily cycle	Few suitable sites; ecosystem impacts
Wave	20–35%	200–400	Vast resource	Immature; harsh marine environment

Intermittency and Storage

Wind and solar are *variable* (output depends on weather) and *non-dispatchable* (cannot be ramped at will). A decarbonised grid therefore needs **flexibility** via: (i) lithium-ion battery storage (expanding rapidly: Hornsdale, South Australia; Moss Landing, California); (ii) pumped hydro (Dinorwig, Wales; still ~95% of global storage capacity); (iii) demand-side response and smart grids; (iv) interconnectors pooling weather variability across geography (North Sea Link 720 km UK–Norway cable, 2021); (v) long-duration storage — green hydrogen, compressed air, flow batteries, thermal.

Exam tip — capacity factor vs installed capacity: Do not confuse installed capacity (GW, a nameplate rating) with actual generation (GWh). A 1 GW solar farm with 20% capacity factor generates less than a 0.5 GW nuclear plant with 90% capacity factor. Examiners reward candidates who use *capacity factor* correctly when comparing technologies.

9. Nuclear Power

Nuclear fission is a low-carbon, high-capacity-factor, energy-dense technology but carries social and economic baggage. Globally ~9% of electricity, concentrated in France, USA, China, South Korea, Russia, Japan.

- **Life-cycle emissions** ~12 g CO₂/kWh — comparable with wind.
- **Capital costs** are very high: Hinkley Point C (UK EPR, 3.2 GW) cost ~£46 bn and is a decade late; Vogtle 3 & 4 (USA) ~\$35 bn.
- **High-level waste** remains radioactive for ~100,000 years; Finland's Onkalo deep geological repository is the first operational.

- **Accident risks:** Chernobyl 1986, Fukushima 2011. Fukushima triggered Germany's nuclear phase-out and Japan's temporary shutdown, adding ~1 GtCO₂ to global emissions in the decade following.
- **Decommissioning costs** are large and long-tailed (UK's Sellafield legacy ~£130 bn).
- **Small Modular Reactors (SMRs)** — factory-built, 50–300 MW units — are being developed (Rolls-Royce UK, NuScale USA, BWRX-300) as a potentially cheaper, faster option.

10. Energy Transition and Decarbonisation

An **energy transition** is a structural shift from one dominant energy system to another (wood → coal; coal → oil & gas; fossil → low-carbon). The current transition is the first deliberately *policy-driven* transition in history, responding to climate science rather than new resource discoveries.

Decarbonisation: The process of reducing the carbon intensity of an economy — the ratio of CO₂ emissions per unit GDP — ultimately toward near-zero ("net zero"). Combines supply-side change (renewables, nuclear, CCS) with demand-side change (efficiency, electrification, behavioural shift, circular economy).

Net zero: A state in which anthropogenic greenhouse-gas emissions are balanced by anthropogenic removals over a specified timeframe, producing no net addition to the atmosphere. Remaining residual emissions (aviation, agriculture, industry) are offset by natural sinks (afforestation, soils) or engineered removals (BECCS, DACCS).

International Policy Architecture

- **UNFCCC 1992** — Rio Earth Summit framework convention.
- **Kyoto Protocol 1997** — first legally binding targets, developed countries only.
- **Paris Agreement 2015 (COP21)** — legally binding, universal framework with three goals: (i) limit warming to *well below* 2°C, pursuing 1.5°C; (ii) adaptation and resilience; (iii) align finance with low-carbon development. Each country submits **Nationally Determined Contributions (NDCs)**, updated every five years, with a ratchet mechanism.
- **Glasgow Climate Pact 2021 (COP26)** — first explicit mention of coal "phase-down"; methane pledge; deforestation declaration.

- **COP27 Sharm el-Sheikh 2022** — historic agreement on a *Loss and Damage fund*.
- **COP28 Dubai 2023** — first Global Stocktake; "transition away from fossil fuels" wording (not "phase-out"); tripling renewables, doubling efficiency by 2030.

Nationally Determined Contribution (NDC): A country's self-defined pledge under the Paris Agreement setting out its mitigation and adaptation actions. NDCs are submitted every five years and should reflect the highest possible ambition. Current aggregated NDCs imply ~2.5–2.9°C warming by 2100 — far short of 1.5°C.

Net-Zero Timetables

- **UK** — net zero by 2050 (2008 Climate Change Act; amended 2019). Carbon budgets set by Climate Change Committee.
- **EU** — net zero by 2050 (European Climate Law 2021); –55% by 2030 vs 1990 ("Fit for 55"); REPowerEU 2022.
- **USA** — net zero by 2050 (Biden 2021); Inflation Reduction Act 2022 = ~\$370 bn clean-energy subsidy. 2025 policy direction uncertain under Trump 2.0.
- **China** — net zero by 2060; peak emissions before 2030.
- **India** — net zero by 2070; ambitious 500 GW non-fossil by 2030.
- **IEA Net Zero by 2050 scenario** — no new oil, gas or coal field approvals; triple renewables; double efficiency; EVs > 60% of new car sales by 2030.

Carbon Pricing

Putting a price on CO₂ internalises the climate externality:

- **EU Emissions Trading System (ETS):** world's largest cap-and-trade market since 2005; prices reached €100/tCO₂ in 2023.
- **UK ETS:** post-Brexit national scheme launched 2021.
- **Carbon Border Adjustment Mechanism (CBAM):** EU levy on imported steel, cement, aluminium, fertiliser, electricity, hydrogen (transitional from 2023, full from 2026) — prevents carbon leakage, penalises dirty producers.
- **Carbon taxes:** Sweden (\$130/tCO₂), Canada federal floor (\$80 rising to \$170/tCO₂ by 2030).

Carbon tax: A price levied on each tonne of CO₂ (or CO₂-equivalent) emitted, intended to internalise the social cost of carbon and incentivise low-carbon choices. Contrasts with *cap-and-trade* (fixes quantity, lets market set price).

Stranded asset: An asset that suffers unanticipated write-downs, devaluations or conversion to liabilities as a result of changing market, regulatory or environmental conditions. In the energy transition, stranded assets include fossil-fuel reserves that cannot be burned within a 1.5°C carbon budget (estimated ~60–80% of proven reserves) and the associated infrastructure (pipelines, power stations, refineries).

Exam tip — use the "unburnable carbon" argument: Showing awareness that current proven fossil reserves exceed the remaining 1.5°C carbon budget by a factor of 3–4 lets candidates link physical geography (carbon cycle) to economic geography (stranded assets) — synoptic thinking that scores at the top band.

11. The Hydrogen Economy

Hydrogen is an energy *carrier*, not a primary source — it must be manufactured from something else. Its appeal is that it burns or oxidises (in a fuel cell) to produce only water, and is dense enough for uses where batteries struggle: shipping, aviation, steel, heating.

Hydrogen colours: Industry jargon for H₂ by production route. *Grey hydrogen* is made from natural gas via steam methane reforming (SMR), emitting ~10 tCO₂/tH₂ — not low-carbon. *Blue hydrogen* is grey + CCS, capturing 60–95% of emissions. *Green hydrogen* is electrolysis powered by renewables — near-zero emissions but energetically expensive. Also *pink* (nuclear-electrolysis) and *turquoise* (methane pyrolysis).

Green hydrogen cost has fallen from ~\$6/kg in 2020 to ~\$3–5/kg in 2024; grey is ~\$1–2/kg. Parity depends on cheap renewables and carbon pricing. Early projects: NEOM Green Hydrogen (Saudi Arabia, 4 GW, online 2026); HyBrit (Sweden) green-steel pilot; Teesside H₂ cluster (UK).

Potential uses:

- **Steel:** replacing coking coal in direct reduction. Carbon-intensive sector (~8% global CO₂).
- **Cement:** high-temperature heat for kilns (though process emissions still need CCS).
- **Fertiliser:** ammonia synthesis currently uses grey H₂; decarbonising it fixes ~1% of global CO₂.
- **Heavy transport:** HGVs, ships (ammonia), aircraft (sustainable aviation fuel feedstock).
- **Long-duration storage:** seasonal storage of summer solar surplus for winter heating.

Limits: electrolyser efficiency (~70%) means a green H₂ heating system uses ~5× the electricity of a heat pump for the same warmth; H₂ leaks are themselves an indirect GHG (extends atmospheric CH₄ lifetime).

Exam tip — hydrogen is not a silver bullet: Top-band answers recognise that H₂ is most useful in *hard-to-electrify* sectors (steel, shipping) and less compelling where direct electrification works (cars, domestic heating). Avoid the "H₂ will solve everything" framing and apply it selectively.

12. Carbon Capture, Utilisation and Storage (CCUS)

Carbon capture and storage (CCS): A suite of technologies that separate CO₂ from large industrial or power-plant flue streams (or, in direct air capture, from ambient air), compress it, transport it by pipeline or ship, and inject it permanently into deep saline aquifers, depleted oil reservoirs or basalt formations. Variants include CCUS (where captured CO₂ is utilised in products) and BECCS (bioenergy with CCS, giving net-negative emissions).

Three capture routes:

- **Post-combustion:** amine scrubbing of flue gas after normal combustion. Retrofittable. Most deployed form.
- **Pre-combustion:** gasify fuel to syngas (CO + H₂), shift to CO₂ + H₂, separate before burning.
- **Oxy-fuel:** burn in pure O₂ producing concentrated CO₂/H₂O flue.

Storage options: depleted oil and gas fields (Sleipner Norway since 1996, 20 MtCO₂ injected; Weyburn Canada); saline aquifers (Gorgon Australia; Northern Lights Norway 2024); basaltic mineralisation (Carbfix Iceland, injected CO₂ forms carbonate rock in months to years).

Global CCS capacity is ~50 MtCO₂/yr in 2024 — a tiny ~0.1% of global emissions. The IEA Net Zero pathway requires ~6 GtCO₂/yr of CCS by 2050, a 120-fold scale-up. UK policy bets on industrial clusters: HyNet (North West), East Coast Cluster (Teesside/Humber), Acorn (Scotland).

Critics argue CCS: (i) lengthens fossil-fuel lock-in by offering a "mitigation deterrence" excuse; (ii) consumes ~20–30% of a plant's energy output, raising costs; (iii) most captured CO₂ to date has been used for *Enhanced Oil Recovery* — producing more oil. Defenders respond that hard-to-abate sectors (cement, steel) have no other route to zero, and BECCS/DACCS are required in almost all 1.5°C pathways.

BECCS — Bioenergy with Carbon Capture and Storage: Growing biomass that absorbs atmospheric CO₂, combusting it for energy, then capturing and burying the resulting CO₂. In principle gives *negative* net emissions. Critiqued for land-use competition (food, biodiversity) and life-cycle accounting; central to most IPCC 1.5°C scenarios yet barely deployed.

13. Mitigation and Adaptation

Climate policy operates through two distinct but complementary strategies:

- **Mitigation:** reducing emissions or increasing sinks, addressing the *cause* of climate change. Tools: renewables, nuclear, efficiency, electrification, behaviour change, reforestation, CCS, carbon pricing.
- **Adaptation:** adjusting to present and projected impacts, addressing the *consequences*. Tools: flood defences, drought-tolerant crops, managed realignment, early-warning systems, planned relocation, air-conditioned cooling centres.
- **Loss and Damage:** a third category recognising that some impacts (sea-level rise swallowing small island states; heat beyond adaptation limits) cannot be adapted to. COP27 Sharm el-Sheikh 2022 agreed to establish a dedicated fund; COP28 operationalised it with pledges (>\$700 m) that remain far below the scale of need.

Mitigation and adaptation interact: insufficient mitigation forces more costly adaptation; some adaptation (air conditioning, desalination) itself consumes energy and, if fossil-powered, worsens mitigation.

14. Required Case Studies

Case study 1 — Amazon rainforest carbon loss: The Amazon holds ~150–200 GtC in biomass and soil, ~10% of global terrestrial carbon. *Gatti et al. (2021, Nature)* used nine years of aircraft measurements to show that the *south-eastern Amazon (especially Pará and Mato Grosso)* is now a *net source of CO₂* rather than a sink, driven by deforestation for cattle ranching and soy, plus escalating dry-season fires. 2019 — the first Bolsonaro year — saw deforestation jump ~34% and an estimated 0.5 GtCO₂ released from fires. Bolsonaro (2019–2022) weakened IBAMA (enforcement agency), gutted Indigenous demarcations, and publicly encouraged frontier settlement. President Lula (inaugurated January 2023) pledged zero deforestation by 2030, restored ministries, and in 2023 deforestation fell ~50%. Policy mechanisms include: *REDD+* (Reducing Emissions from Deforestation and Degradation), paying tropical states to keep forests standing — Amazon Fund (Norway, Germany donors); Indigenous land rights — Indigenous territories lose forest at ~1/3 the rate of surrounding land. Tipping-point concern: beyond ~20–25% cumulative deforestation, the eastern Amazon's internal water recycling (50% of rainfall is recycled from ET) may collapse, flipping the biome to savanna and releasing ~50–100 GtC.

Case study 2 — Germany's Energiewende: Germany's *energy transition* is the most ambitious decarbonisation in a major industrial economy. Origins: 1970s anti-nuclear movement + 1986 Chernobyl. Key milestones: 2000 Renewable Energy Sources Act (EEG) introduced feed-in tariffs that drove solar-PV deployment and global cost decline; 2011 nuclear phase-out decision after Fukushima; 2019 Climate Action Law (net zero by 2045 after 2021 constitutional court ruling); coal phase-out by 2038 (under negotiation to 2030); final three nuclear plants closed April 2023. Achievements: renewables rose from ~6% of electricity in 2000 to ~55% in 2024; wind and solar dominate. Challenges: (i) nuclear closure forced continued reliance on lignite (coal and lignite still supplied about a quarter of electricity); (ii) Nord Stream 1 & 2 left Germany 55% dependent on Russian gas pre-2022 — Putin's invasion of Ukraine (Feb 2022) triggered a crisis, LNG imports from US/Qatar, rapid LNG terminal construction at Wilhelmshaven/Brunsbüttel; (iii) grid congestion — wind in the north, demand in the south, SuedLink HVDC line under construction; (iv) industrial competitiveness (BASF, chemicals) pressured by gas prices. *REPowerEU* (May 2022) accelerated renewables targets to 45% of EU energy by 2030 in direct response. Germany's Energiewende illustrates both the possibilities and limits of state-led transitions.

Case study 3 — Saudi Arabia and Kuwait: oil-dependent rentier economies: The Gulf oil monarchies exemplify *rentier states* — governments funded by resource rents rather than taxation,

shaping political economy. Saudi Arabia holds ~17% of proven oil reserves, Kuwait ~6%. Oil and gas provide ~70% of Saudi fiscal revenue and ~80% of Kuwait's, making them acutely exposed to energy transition. **Saudi Vision 2030** (2016, Mohammed bin Salman) aims to diversify via tourism, entertainment, renewables and mega-projects: *NEOM* — a 170 km linear city "The Line" in the Tabuk desert, with Oxagon (industrial), Trojena (ski resort); progress slower than announced, budget cut 2024. *Aramco IPO* (December 2019) listed 1.5% of Saudi Aramco for \$29.4 bn — largest ever IPO; valued the company at ~\$1.7 trillion. **Renewables paradox:** Saudi has announced 50% renewable electricity by 2030 and the NEOM Green Hydrogen Project (4 GW electrolyser, 600 t/day H₂). Yet simultaneously Aramco plans to expand capacity and Saudi Arabia was central to weakening COP28 fossil-fuel language. Kuwait remains more conservative: no Vision-style plan; gas burned for summer air-con; welfare-state entitlements vulnerable if oil revenues collapse. Both illustrate the tension between short-term rents and long-term stranded-asset risk: ~\$1–2 trillion of Gulf oil reserves may be "unburnable" within a 1.5°C carbon budget.

Case study 4 — Drax power station, North Yorkshire (UK): Drax is western Europe's largest power station (original coal capacity 3.9 GW). Between 2012 and 2018 four of its six units were converted from coal to *biomass* — compressed wood pellets, mostly imported from the south-eastern USA (Louisiana, Mississippi, Georgia, Arkansas) and British Columbia. Drax supplies ~4% of UK electricity and is officially counted as "renewable" under UK Renewable Obligation Certificates (ROCs) and Contracts for Difference (CfDs), receiving ~£600 m/yr in subsidies.

Critics' case: (i) wood pellet supply chains release ~200–300 kg CO₂/MWh from harvesting, drying (with natural gas), and trans-Atlantic shipping; (ii) accounting treats smokestack emissions as zero because tree regrowth will reabsorb the carbon — but regrowth takes 40–100 years, breaching the carbon-budget window; (iii) *Enviva*, Drax's largest pellet supplier, went bankrupt in 2024 amid investigations into sourcing from mature hardwood forests (not just residues); (iv) a 2022 BBC Panorama and 2024 Ofgem investigation found Drax had overstated sustainability compliance. **Drax's response:** stricter FSC sourcing, planned BECCS retrofit (capturing ~8 MtCO₂/yr) to make Drax a net-negative site by 2030 — £2 bn project pending government support. Drax is a live controversy about what counts as "renewable" and how accounting loopholes can disguise continued combustion emissions.

Case study 5 — Iceland: geothermal leadership and carbon mineralisation: Iceland's ~380,000 people enjoy ~100% renewable electricity: ~70% hydro, ~30% geothermal, on a grid

isolated from Europe. The island sits on the Mid-Atlantic Ridge, giving shallow high-temperature geothermal resources. Major plants: *Hellisheiði* (near Reykjavík, 303 MW electrical + 400 MW thermal district heating); *Námafjall*; *Reykjanes*. Geothermal water (80–95°C) also heats 90% of Icelandic homes via district networks, displacing oil burners that dominated pre-1970. This extraordinarily low-carbon electricity has attracted energy-intensive industry: three aluminium smelters consume ~70% of Iceland's electricity; since 2015 cryptocurrency mining and data centres have arrived. **Carbfix / Climeworks:** at the Hellisheiði plant, Swiss company Climeworks operates *Orca* (4,000 tCO₂/yr, 2021) and *Mammoth* (36,000 tCO₂/yr, 2024) direct air capture plants. The captured CO₂ is mixed with water and injected into porous basaltic lava ~800 m below the surface, where it reacts with Mg/Ca/Fe ions to form carbonate minerals — turning CO₂ into stone in under two years. Scale is tiny compared with global emissions, but the technology is the first commercial permanent carbon mineralisation and a model for replication wherever basalt outcrops coastally (Oman, Indian Deccan traps, Scottish west coast). Iceland also exports know-how to Kenya's Rift Valley geothermal build-out.

14b. Comparing Fossil and Renewable Energy

Dimension	Fossil fuels	Renewables
Energy density	Very high (coal ~24 MJ/kg; petrol ~46 MJ/kg)	Diffuse (solar ~200 W/m ² average, ~1,000 W/m ² peak)
Dispatchability	On-demand (coal, gas); flexible	Variable (wind, solar); hydro dispatchable
Carbon emissions	High (800–1000 g CO ₂ /kWh coal)	Low (4–50 g CO ₂ /kWh life-cycle)
Air pollution	SO ₂ , NO _x , PM, mercury	Negligible in operation
Water use	High (cooling coal/gas ~2 L/kWh)	Low (wind, solar PV); high (CSP, hydro)
Capital intensity	Moderate–high	High upfront, near-zero fuel
Marginal cost	Fuel-dominated	Near zero

Geopolitical reach	Concentrated (OPEC, Russia, Gulf)	Widely distributed; minerals concentrated
Land use	Point source but mining footprint	Area-extensive (wind, solar, biomass)
Scalability in 2024	Mature, plateauing	Accelerating; price declines

15. Arctic Oil and Gas Exploration

The Arctic is warming ~4× faster than the global average, and sea-ice retreat is opening access to a region that the US Geological Survey estimates holds ~13% of undiscovered oil and ~30% of undiscovered natural gas. Major players:

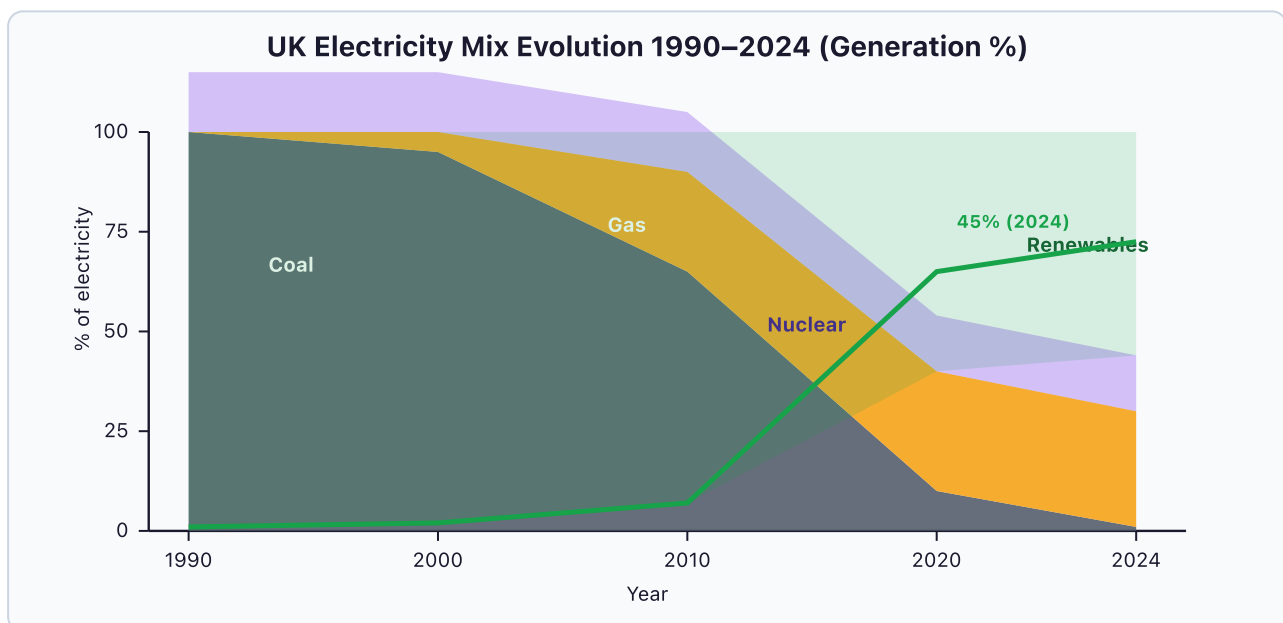
- **Russia** dominates Arctic hydrocarbon activity: Yamal LNG (Novatek, operational 2017, 17.4 Mt/yr); Arctic LNG 2 (sanctioned by US 2023 after Ukraine invasion, stalled); Prirazlomnaya offshore oil platform; the Northern Sea Route being developed as an Arctic shipping lane.
- **USA**: Alaska North Slope (Prudhoe Bay); Willow Project (ConocoPhillips, approved by Biden 2023 despite climate criticism; 180,000 bpd).
- **Norway**: Barents Sea fields (Goliat, Johan Castberg — first oil 2025).
- **Canada, Greenland (Denmark)**: more cautious; Greenland banned new oil exploration 2021.

Geopolitical tension: eight-member Arctic Council (Russia chaired 2021–2023) effectively suspended after Ukraine invasion; Russia–China energy partnership deepening. Environmental concerns: oil spill in sea ice cannot be recovered (no established clean-up method); disruption of Indigenous (Sami, Inuit, Nenets) lifeways; threats to polar bears, walrus, beluga; black carbon deposition on snow accelerates melt. Ethical dilemma: should the last pristine frontier be opened precisely when climate science says most fossil fuels must remain unburned?

16. Synthesis — Linking Carbon Cycle and Energy Security

The two halves of Topic 2.2 converge on a single question: *can a 10-billion-person industrial civilisation sustain itself without continuing to move geological carbon into the atmosphere?* The answer depends on:

1. **Rate of energy transition:** fast enough to stay within the ~400 GtCO₂ remaining 1.5°C carbon budget (as of 2024, ~9 years at current rates).
2. **Protection of natural sinks:** avoiding Amazon die-back, permafrost thaw and ocean-sink weakening that could release hundreds of GtC.
3. **Scaling negative emissions:** BECCS, DACCS, ocean alkalinity, afforestation — needed to offset hard-to-abate residuals and to draw down overshoot.
4. **Justice:** historical emitters (Global North, ~79% of cumulative since 1850) carry responsibility for finance, technology transfer and loss-and-damage commitments to the Global South.
5. **Geopolitical re-ordering:** as oil rents decline, new minerals (lithium, cobalt, copper, rare earths) rise; 2020s–2030s will see strategic competition over Chile's lithium triangle, DRC's cobalt, Indonesia's nickel, Greenland's rare earths.



16b. Critical Minerals and the Transition's New Dependencies

Decarbonisation is not "dematerialisation": a gigawatt of offshore wind needs ~8,000 t of steel, ~5,000 t of concrete, 110 t of copper, 5 t of rare-earth elements; a Tesla Model 3 battery contains ~10 kg lithium, ~8 kg cobalt, ~40 kg nickel. The World Bank estimates mineral demand for the transition will rise 4–6× by 2050. Key minerals and their chokepoints:

- **Lithium** — Chile (Atacama), Australia, Argentina, China; 77% of global processing is Chinese.

- **Cobalt** — Democratic Republic of Congo (~70% of world supply), often from artisanal mining with child-labour concerns.
- **Copper** — Chile, Peru, DRC; demand to double by 2035.
- **Nickel** — Indonesia, Philippines, Russia; Indonesian laterite nickel smelting is coal-powered, creating a "dirty-battery" paradox.
- **Rare-earth elements (Nd, Dy, Tb)** — China ~60% of mining, ~90% of processing; used in permanent magnets for wind turbines and EV motors.
- **Graphite, manganese, silicon, silver** — also strategically important.

This reshapes geopolitics: the USA's Inflation Reduction Act (2022) and the EU's Critical Raw Materials Act (2024) seek "friend-shoring" to break Chinese dominance; Chile and Indonesia have imposed export restrictions to capture more value-added; conflict over DRC's cobalt mines continues; Greenland's rare-earth deposits became a geopolitical flashpoint in 2025. The transition from *fuel* geopolitics to *mineral* geopolitics is one of the defining shifts of the 21st-century world economy.

Exam tip — transition ≠ frictionless: Synoptic essays benefit from acknowledging that the low-carbon transition creates its *own* environmental and geopolitical footprints — deep-sea mining, lithium-triangle water stress, DRC labour abuses, new mineral cartels. This avoids naive "renewables good / fossil bad" framing and hits AO3 evaluation.

17. Case Study Summary Table

Case study	Location	Issue illustrated	Key numbers
Amazon carbon loss	Brazil, esp. Pará, Mato Grosso	Biosphere C sink becoming source	SE Amazon ~+0.5 GtCO ₂ /yr; ~20–25% deforestation tipping point
Germany Energiewende	Germany	Ambitious transition in industrial economy	Renewables 6% (2000) → 55% (2024); nuclear phase-out 2023
Saudi / Kuwait oil dependency	Gulf states	Rentier economies & stranded-asset risk	~70% (KSA) / 80% (KW) fiscal revenue from oil; NEOM >\$500 bn planned

Drax biomass	North Yorkshire, UK	"Renewable" accounting loopholes	~4% UK electricity; ~£600 m/yr subsidy; 8 Mt pellets/yr imported
Iceland geothermal + Carbfix	Iceland	High-renewable grid + carbon mineralisation	~100% renewable electricity; Mammoth DAC 36,000 t/yr (2024)

18. Exam Technique — Extended Response

Eduqas Component 2 tests this content in both short-answer AO1 questions (definitions, processes, quantified fluxes) and longer AO2/AO3 evaluative essays (typically 9, 12, 18 or 20 marks).

Anatomy of a Top-Band 20-Mark Essay

- 1. Introduction (1 paragraph):** define key terms, state thesis, outline structure. Example: "This essay argues that the global carbon cycle is undergoing unprecedented disruption primarily through fossil-fuel combustion, but that regional variations in energy transition — illustrated by contrasts between Germany's Energiewende and Saudi Arabia's rentier model — will determine the rate of future recovery."
- 2. Three developed paragraphs (AO1 + AO2):** each with a topic sentence, case-study evidence (named places, quantified), theoretical framing (e.g. feedback loop, tipping point, rentier state), and mini-evaluation.
- 3. One counter-argument paragraph (AO3):** e.g. "However, CO₂ fertilisation and expanded ocean sinks partially offset anthropogenic emissions, and arguments that the transition is 'already happening' ignore the ~80% share of fossil fuels in primary energy."
- 4. Conclusion:** resolve the thesis, make a judgement with justification, look forward (COP process, NDC ratchets, 2030/2050 deadlines).

Exam tip — triangulate scales: The Eduqas spec rewards the "local–national–global" scale-triangulation approach. A single paragraph covering the Amazon, Brazilian federal policy, and COP global governance of forests scores higher than three paragraphs each confined to one scale. Explicitly move between scales.

Exam tip — quantify every claim: The single most reliable way to raise essay marks is to embed numbers — 120 GtC/yr photosynthesis, 10 GtC/yr fossil emissions, 425 ppm CO₂, ~1.3°C warming to date, 50% of UK electricity from renewables in 2024. Examiners mark-hunt for data.

Sample Questions by Command Word

- **Define** (AO1, 2–3 marks): "Define the term *biological pump*." — give a one-sentence technical definition plus one extension (e.g. mechanism or scale).
- **Describe** (AO1, 4–6 marks): "Describe the main stores of the global carbon cycle." — name, locate, quantify; use a table mentally.
- **Explain** (AO1/AO2, 6–9 marks): "Explain how human activity has disrupted the carbon cycle since 1750." — identify mechanisms (combustion, LUC, cement), quantify fluxes, link to atmospheric rise.
- **Assess** (AO2/AO3, 12 marks): "Assess the extent to which renewable energy can replace fossil fuels in delivering energy security." — balanced argument with named case studies, criteria for judgement.
- **Evaluate** (AO3, 18–20 marks): "Evaluate the view that feedbacks in the carbon cycle make it impossible to limit warming to 1.5°C." — define, weigh evidence, distinguish positive/negative feedbacks, examine the 1.5°C pathway evidence (IEA/IPCC), reach judgement with justification.
- **Discuss** (AO2/AO3, 18–20 marks): "Discuss the contrasting pathways of energy transition in Germany and Saudi Arabia." — compare/contrast, multiple perspectives, reach reasoned synthesis.

Exam tip — read the command word first: "Describe" expects content-only; "Explain" demands mechanisms; "Assess/Evaluate/Discuss" require explicit evaluative paragraphs and a judgement. Mis-reading the command word is the single most common cause of under-performing on long-answer Eduqas questions.

19. Key Terms Glossary

Term	Definition
Carbon cycle	Biogeochemical circulation of C between lithosphere, hydrosphere, atmosphere, biosphere

GtC / PgC	Gigatonnes / petagrams of carbon (1 GtC = 1 PgC = 3.67 GtCO ₂)
Photosynthesis	$6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ (solar energy)
Respiration	Reverse of photosynthesis; organic matter oxidised to CO ₂
Biological pump	Ocean process sinking organic C from surface to sediments
Solubility pump	CO ₂ dissolving in cold polar waters that sink
Ocean acidification	Fall in ocean pH from CO ₂ uptake
Feedback loop	System response amplifying (+) or dampening (–) original change
Tipping point	Threshold beyond which change becomes self-sustaining and irreversible
Energy security	Uninterrupted availability of affordable energy
Energy mix	Combination of primary energy sources in a country/region
Primary vs final energy	Raw form vs delivered form to end-user
Capacity factor	Actual output / theoretical maximum output (%)
Fracking	Hydraulic fracturing to extract shale gas/oil
Decarbonisation	Reducing carbon intensity of an economy toward zero
Net zero	Emissions balanced by removals; no net atmospheric addition
NDC	Nationally Determined Contribution (Paris Agreement pledge)
ETS	Emissions Trading System (cap-and-trade market)
CBAM	Carbon Border Adjustment Mechanism (EU 2023–)
Carbon tax	Price per tonne CO ₂ emitted

Stranded asset	Asset devalued by transition (e.g. unburnable fossil reserve)
CCS / CCUS	Carbon Capture and (Utilisation and) Storage
BECCS	Bioenergy with CCS — potentially negative emissions
DACCS	Direct Air Carbon Capture and Storage
Hydrogen colours	Grey (gas), Blue (gas+CCS), Green (electrolysis+renewables), Pink (nuclear), Turquoise (pyrolysis)
Mitigation vs Adaptation	Tackling cause vs managing consequence of climate change
Loss and Damage	Climate impacts beyond adaptation limits; COP27 fund
REDD+	Reducing Emissions from Deforestation and Degradation plus conservation
Rentier state	Economy funded by resource rents rather than taxation
Energiewende	Germany's energy transition from fossils/nuclear to renewables

Common Confusions to Avoid

- **GtC vs GtCO₂**: always check units — 1 GtC = 3.67 GtCO₂. Emissions are commonly reported in CO₂; stores in C.
- **ppm vs ppb**: CO₂ is ppm (parts per million, 425); CH₄ is ppb (parts per billion, 1,930).
- **Climate change vs weather**: climate is the statistics of weather over ≥30 years; individual heatwaves are weather events with climate-change-amplified probability.
- **Renewable vs low-carbon**: nuclear is low-carbon but not renewable; biomass is renewable but not necessarily low-carbon.
- **Carbon neutral vs net zero**: used interchangeably but "net zero" typically covers all GHGs and includes Scope 3; "carbon neutral" often narrower.
- **Capacity vs generation**: 1 GW nuclear ≠ 1 GW solar in annual output (see capacity factor tip).

20. Summary — What You Must Know for Eduqas 2.2

- The **global carbon cycle** with the eight major stores and the key natural and anthropogenic fluxes quantified in GtC/yr.
- The contrast between the **fast biological** and **slow geological** cycles, and why anthropogenic combustion short-circuits between them.
- The **biological and solubility pumps** and their role in deep-ocean carbon storage.
- **Keeling curve** 315 → 425+ ppm and **ocean acidification** –0.1 pH.
- Positive (permafrost, Amazon, ice-albedo, wildfire, ocean outgassing) and negative (CO₂ fertilisation, silicate weathering, ocean uptake) feedbacks; named tipping points.
- Definitions and components of **energy security**, **energy mix**, and contrasting national profiles (UK, Germany, France, Norway, Saudi, China, USA).
- Conventional vs unconventional fossil fuels; renewables' strengths and intermittency problem; nuclear's trade-offs.
- The **Paris Agreement**, NDCs, net-zero timelines, carbon pricing (ETS, CBAM, carbon tax).
- The **hydrogen economy** (colours, uses, limits) and **CCS/BECCS/DACCS**.
- **Five case studies**: Amazon, Germany Energiewende, Saudi/Kuwait, Drax, Iceland (Hellisheiði + Carbfix).
- Arctic oil and gas exploration and the ethical contradiction with climate commitments.
- Synthesis: the transition must happen fast enough to stay within the ~400 GtCO₂ budget, while protecting natural sinks and scaling negative emissions — with justice implications for the Global South.
- Critical minerals as the "new oil": lithium, cobalt, copper, nickel, rare earths shape the geopolitics of the transition.
- Interconnections: the carbon cycle and energy security are not two topics but one — the global climate is the integrated consequence of billions of energy decisions.

This resource covers Eduqas A-Level Geography (A110QS) Topic 2.2 at full A-level depth, with integrated quantitative data, five required case studies, and essay-writing technique for 20-mark extended responses.